Pilot Tone Trigger

Quick reference — controls & first-run setup

Watches an audio input carrying a continuous pilot tone. While the tone is present the primary source is considered healthy; if the tone drops, the app sends MIDI to the dLive to switch to the backup — an **Action** (Control Change), a **Scene** recall (Program Change), or both. The app can also generate the tone itself (see Tone Generator). Typical tone-drop → MIDI latency with default settings: ≈150–250 ms.

Menu bar icon

Click	Show / hide the control panel.
Right-click	Quit the app (there's no Dock icon or app menu — this is the only way to quit).
Icon colour	The sine-wave mark: grey = stopped. Blue = running, tone present. Red = tone lost — failover sent. Amber = tone restored but latched on backup (press Reset). Purple = level fine but the tone looks corrupted (Signal Integrity).

Connection

Choose one. Both reach the console the same way underneath — pick whichever needs less new software.

Direct TCP	Opens its own connection to the console. Host must be the **MixRack's IP address** — never the Surface's — regardless of which unit your network cable is physically plugged into (the Surface bridges through to the MixRack). Default: **192.168.1.70** (the MixRack factory IP); Port defaults to 51325.
Existing MIDI Port	Sends over a MIDI port already routed to the console — most easily the free **dLive MIDI Bridge**, or A&H's driver / loopMIDI / IAC. Pick it from the dropdown. When the port is dLive MIDI Bridge, this app identifies itself by name (“Pilot Tone Trigger”) in the bridge's MIDI Monitor instead of showing as a generic connection.
Base MIDI Channel	Must match Utility → Control → MIDI on the console.
Test Connection	Connects and disconnects once. Green tick = console reachable; red = fix IP/network. Proves reachability, not the console-side mapping — verify that end-to-end once.

Audio Input

Device	The audio interface carrying the pilot tone.
Channel	Which input channel on that device to listen to.

Tone Generator

If Signal Integrity (below) is on, it's only as reliable as this reference tone: an EXTERNAL tone generator can drift a little in frequency or carry more inherent distortion than this app's own synthesized sine, which eats into how tight the Purity Threshold can be set before it false-trips on a perfectly healthy signal.

What it's for	The app can be both ends of the chain: generate the pilot tone to an output, route it through the system under watch (e.g. out to the console and back), and loop it into the Audio Input. If the loop breaks anywhere along the path, the tone disappears and the failover fires.
Output Device	Which audio output carries the generated tone.
Output Channel	Which channel of that device gets the tone — the other channels stay silent. Changing it, or the Output Device, moves a playing tone straight away.
Frequency (1000 Hz)	Sine frequency, 20–20 000 Hz. Retunes live, click-free.
Level (–20 dBFS)	Output level. Changes ramp smoothly.
Start / Stop Tone	Independent of the detection engine — the tone keeps running across detection Start/Stop. It never auto-starts on launch; starting it is always a deliberate click.
If the Mac's audio resets	Installing an audio driver, or sleep/wake, resets the Mac's audio system. The app notices within about 2 seconds and reopens the tone and the input on the devices you chose (Latch kept, nothing sent), and says so. A tone that stops for another reason is reopened within 3 seconds, retried every 10.

MIDI Output

What each state change sends. Columns in the app: Signal Lost (red, left) and Signal OK (green, right) — each state has its own settings.

Action (CC)	Sends a Control Change per state — its own CC number and value. On the dLive an incoming CC triggers an **ACTION**, e.g. setting a group of channels to Preamp A/B/C/D, toggling an insert on/off, and so on. The console must be told what that CC means: map it under **Surface** → **Control** → **Actions** → **MIDI Recall**, matching the CC number, value and Base MIDI Channel set here. New installs default to **CC 85** — Pilot Tone Trigger's own number in the dylanmaudio suite, value **127** for tone present and **0** for tone lost. Existing configs keep whatever number they already had.
Scene (PC)	Sends a Program Change (with Bank Select) per state — recalls a **SCENE** directly, no mapping needed. Scene numbers 1–500, as shown on the console's Scenes screen.
Both	Fires the Action first, then the Scene, on every state change (and on Reset).

Important — a Scene recall changes everything the Scene stores, not just what you intend. Set each Scene's Recall Filter to allow only the items the failover should touch. For Actions, nothing happens at all until the matching MIDI Recall mapping exists on the console.

Failback

Automatic	When the tone returns, the Signal OK output fires by itself and the system is back on primary. Switching from Latch to Automatic releases an armed latch — and fires the Signal OK output immediately if the tone is already back.
Latch until Reset	After a dropout the app stays on backup even when the tone returns — no automatic flap back onto a source that just failed. While latched, further drop/restore cycles send nothing at all (the console is already on backup) until Reset or a switch to Automatic releases it. Recommended for shows.
Primary / Backup	The bar under Start / Stop shows which side the console is on while the app runs: **On Primary (A)** in blue, **On Backup (B)** in red — the side the last output that reached the console put it on. Right after Start it reads "not switched since Start": Start sends nothing, so the console is wherever it was left until the first switch.
Don't fail over for a Mac audio reset	Off by default. When on, a dropout is held only if macOS reports its audio system restarting (an audio driver install, sleep/wake) just before it — never on silence alone, so a broken loop always fails over at once. Ends when the tone returns (nothing sent); fails over once the audio is back without the tone, or after 5 s at most.
Reset button	The same bar turns amber and becomes **Reset to Primary** once the tone is back while latched. Sends the Signal OK output and re-arms. Refuses if the tone is still below threshold.
Start / Stop Tone (top)	Beside Flip A/B: the tone generator's on/off, same as the button in the Tone Generator section (its settings are there).
Long messages	Hover over the status message to read all of it.
Flip A/B button	Manual override — sends the opposite of the side shown above it, any time the app is running; a Flip to B shows as **Held on Backup** until Reset or Flip. Sends regardless of what Level or Signal Integrity currently think. Flipping to backup engages the same latch a real dropout would (so a healthy reading can't quietly flip it back); flipping to primary clears it, like Reset.

Level meter

Amber line	Threshold — the level must be above this to count as tone present (recover point).
Red line	Trip point (Threshold − Hysteresis) — the level must fall below this to count as tone lost.
Between the lines	State holds; the status caption reads “In hysteresis band” and shows the exact trip level.

Sensitivity

Defaults are tuned for a continuous pilot tone.

Threshold (–45 dB)	Set below the tone level. Raising it closer to the tone speeds up detection.
Dropout Hold (50 ms)	Time below the trip point required before failing over (rides out single glitched audio blocks).
Hysteresis (6 dB)	Gap between recover and trip lines — prevents flapping at the threshold.
Min Retrigger (200 ms)	Minimum time between consecutive switches, even if the level keeps crossing.
Env Release (30 ms)	How fast the level reading falls — the main latency lever. Shorter = faster detection.

Important — watching programme material instead of a tone? Raise Dropout Hold above the material's longest natural silence and lengthen Env Release (~300 ms), or quiet passages will trip a failover.

Signal Integrity (optional)

Off by default. Only meaningful watching a tone this app generated, or an external source at a precisely known, stable frequency — see the Tone Generator note above.

What it's for	Level alone can't tell a healthy tone from noise or a garbled signal at the same volume. This checks how much of the incoming audio actually sits at the tone's own frequency — clicks, dropouts, noise, clipping and a wrong/garbled tone all show up the same way, as energy that doesn't belong there.
Purity Threshold / Hysteresis / Hold / Min Retrigger	Same shape as Sensitivity above, against a 0–1 purity ratio instead of dB. The starting values are a deliberately conservative placeholder, not a validated default — every interface and signal chain has its own noise floor. Tune to your own rig.
Warn only	Reports Degraded and counts it. Sends nothing — the console keeps running on whatever it's already on.
Also fail over	Additionally fires the same Signal Lost / Signal OK outputs a real dropout would, while still showing Degraded (not Lost) so it's clear why. Recovers on its own once purity clears — no Reset needed, and independent of Failback mode/Latch.

Important — this is not a clone of any hardware vendor's proprietary integrity signal. It's a from-scratch check built for a tone that has to survive an analogue stage and gain changes, unlike some hardware digital-integrity signals which explicitly cannot.

Running it

Start / Stop	Opens (or closes) the audio input and begins (or stops) watching the level. The app arms assuming the tone is present — a feed that is already dead trips the failover after Dropout Hold.
Status tile	Stopped / Signal Present / Signal Lost / Restored — Latched / Signal Degraded (violet, Signal Integrity only). Mirrors the menu bar icon colour. A red “Start Failed” tile names the failing subsystem (console / audio input) in full.
First launch	macOS will prompt for microphone access the first time you click Start — allow it, or the app won't receive any audio.
Settings	Persist to ~/Library/Application Support/Pilot Tone Trigger/config.json.

Bitfocus Companion

Loopback only, port 8772, no setup needed on this side — add this app in the dylanmaudio Companion module.

What it controls	Run, Failback mode, Reset, and the tone generator — as Companion buttons, with this app’s own state shown back on them (works from Companion even if the popover isn’t open).
Allow Companion control	On by default. Off refuses every Companion press; Companion can still show why.
Lock show-critical controls	Off by default. On refuses Run-off and Failback mode from Companion — Reset and the tone generator are never locked.

If something goes wrong

Red: macOS audio stuck	Rare, after macOS's audio system resets (seen when forcing many resets in a row). A macOS Core Audio fault, not the app's. Nothing is sent while it lasts. Quit Pilot Tone Trigger and open it again.
Red: Not reaching the console	An Action or Scene failed to send. The panel stays red until one gets through, and the Primary / Backup bar keeps showing the side the console was last actually switched to — so a failover that never arrived shows as still On Primary.
Export log	Click **Export log** in the panel footer. A diagnostics zip lands on your Desktop and opens in Finder — this session's log, the previous one, and your settings (licence details removed). Send that zip with your report; nothing leaves your Mac unless you send it.
What the log holds	Your settings at launch and each change, the audio and MIDI devices present, every Start and Stop, every failover and signal-OK sent to the console (and any that failed), the latch, each Reset and Flip, every error the panel showed you, your licence or trial state. Every launch is logged automatically.
The log folder	Right-click the menu bar icon and choose **Open Log Folder** (**~/Library/Logs/dylanmaudio/Pilot Tone Trigger/**, newest 20 launches kept).
Send feedback	Also in the footer — a short message straight to me, no email client needed.

What's new in v1.2.6

• Fixed: after the Mac's audio system reset (an audio driver install, or sleep and wake), Pilot Tone Trigger could fail to find a Dante Virtual Soundcard, or another device whose driver comes back a few seconds after the Mac's own, and kept retrying without ever finding it: the tone stayed off and the input dead until you pressed Rescan or relaunched. It now looks for the device again until it is back.
• Fixed: once the tone generator came back after an audio-system reset, the panel could go on saying "the pilot tone is NOT playing" until you pressed Start, Stop or Reset. The warning now clears as soon as the tone is playing again.
• Fixed: the built-in tone generator could drop out for a moment about once a second on some Macs — most often older Intel Macs — and again each time the panel opened or Rescan was pressed. Two background jobs (a once-a-second check for an audio-system reset, and listing your audio and MIDI devices) briefly paused the app's audio; both now run without that pause. On those Macs the same pause could also interrupt the audio input while armed. The tone also plays with a larger audio buffer, and the session log records any gaps the Mac reports.